voxring

Voice-to-everything plugin
for MX Master 4 and MX Creative Console.

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MX Creative Console + MX Master4 & Actions Ring Innovate with the Actions SDK

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What VoxRing is?

A Logi Actions SDK plugin that makes the MX Master 4 side button
and Creative Console the control surface for everything you'd normally type.

33 voice-related actions across 8 groups:

Group	What it does
1 Voice	Dictate, Push to Talk, Voice Assistant, Voice Note, Read Back, Replay Last
2 Send	AI-formatted send to 9 destinations, History folder
3 Quick Send	One-tap voice-to-destination for 7 targets
4 Text Tools	AI toggle, Noise Gate, Filler Cleaner, Case Transform, Translate, Append, Edit, Regenerate
5 Controls	Mute Mic, Cancel, Open Voice Notes
6 Settings	Language Toggle (Auto/EN/DE), full Settings panel
7 Meters	Live audio level dial, rolling waveform, word count
8 Utilities	Guitar Tuner (Easter egg)

Why this could only ship on Logi hardware?

VoxRing is not a voice app with a keyboard shortcut bolted on.
Every Logi hardware affordance is load-bearing.

Affordance	What VoxRing does with it
Side button press + release events	Push-to-talk with hybrid short-tap or long-hold detection
Haptic motor	7 mapped events: record start, pulse, auto-stop warning, transcription done, dial tick, send success, send failure
Thumb wheel	Scrolls through history; each step fires a haptic tick
Creative Console button grid	9-destination Send folder; Guitar Tuner control surface
Creative Console displays	Live dB meter, rolling waveform, word count with estimated read time
Logi Options+ sidebar	8 labelled groups, per-action descriptions, consistent Lucide icon set
Web Applications as Actions	Embedded settings panel: mic selector, engine toggle 5 API key fields with test buttons, model status, TTS test

SDK primitives used: **PluginDynamicCommand**, **PluginDynamicFolder**, **PluginDynamicAdjustment**, **ProcessButtonEvent2**, **PluginEvents.AddEvent** + **eventMapping.yaml**, **HttpListener** settings pattern.

The Stack

Offline by default. Cloud / AI optional. Encrypted at rest.

Voice is captured via NAudio at 16 kHz mono, running entirely in a background thread with no UI involvement.
Speech recognition is handled by two offline engines: Vosk at 40 MB for fast results, and Whisper.net Small at 460 MB for high accuracy with accents and mixed languages. The user picks the engine; the plugin switches transparently. Whisper also auto-detects whether the speaker used English or German and flashes the result on the action button for four seconds.

For read-back and Voice Assistant, VoxRing uses Piper, a fully offline neural TTS engine with an English voice (Amy) and a German voice (Thorsten).
Piper is bundled inside the package and auto-installed on first plugin load with no prompt to the user. Windows SAPI serves as a fallback for edge cases where Piper cannot run.

AI formatting is handled through a shared IAiProvider interface.
Any popular AI can be plug in behind that interface, so swapping providers requires zero changes to action code.
Before any text reaches a destination, it passes through a post-processing pipeline: optional filler-word removal, case transform, and AI-powered translation into five languages. Each step is independently toggleable.

Every API key and webhook URL is encrypted at rest using Windows DPAPI with plugin-specific entropy.
Legacy plaintext values are migrated automatically on first read.
The settings UI is a single-file HTML page served by an embedded HttpListener on localhost, which auto-discovers a free port from 8090 upward.
Haptics are wired through eventMapping.yaml with seven distinct events covering every state transition from record start to send failure.

The entire plugin ships as a single .lplug4 file with Piper and its voice models bundled.
No external dependencies, no separate installers.

Shipped, not vaporware

Everything below is live and working today.

All 33 actions are registered across 8 sidebar groups in Logi Options+, each with a Lucide icon and a user-facing description.
All 9 destinations are active: Clipboard, Type Out, Email, Slack, Discord, Teams, Calendar, WhatsApp, Notion, and Telegram.
Messaging apps that lack a full API open via URI handler and pre-paste from clipboard, so they work immediately without any setup.

Voice Assistant is fully implemented.
The user taps to record a question, the AI responds, and Piper reads the answer aloud through the speakers.
Conversation history carries across turns and resets automatically after five minutes of inactivity.
Tapping while Piper is speaking stops it immediately.

What's next

The architecture is extensible by design.

In the near term, the focus is on closing the remaining gaps in destination coverage.
WhatsApp and Telegram already work via clipboard bridge and URI handler. The next step is native API integration for both, plus Notion's REST API to move beyond the current clipboard fallback. Email contact resolution is planned: saying "email Anna about the meeting" would resolve the first name to an address from local contacts and pre-fill the mailto link. A Whisper Medium model option is also on the list for users who want the highest possible accuracy and have the disk space for it.

On the longer roadmap, macOS parity is achievable by swapping NAudio for a macOS-compatible audio capture library and replacing SAPI with Piper's macOS binary.
The plugin is a strong candidate for Logitech Marketplace submission, which would bring it to the full MX Creative Console user base with localized descriptions and screenshots. A custom wake-word feature would let users start recording without pressing any button, using an always-on low-power voice activity detector.
Finally, calendar write-back via Google Calendar or Microsoft Graph would turn a spoken sentence into a real calendar event, not just a URL.